Higher derivatives

1 Suppose $y = \frac{2}{x^5}$. Find:

a $\frac{dy}{dx}$

b $\frac{d^2y}{dx^2}$

c $\frac{d^3y}{dx^3}$

2 Suppose $y = 5x^8$. Find:

a $\frac{dy}{dx}$

b $\frac{d^2y}{dx^2}$

c $\frac{d^3y}{dx^3}$

d $\frac{d^4y}{dx^4}$

e $\frac{d^5y}{dx^5}$

3 Consider the function $y = -2x^4 + 4x^3 - 7x^2 + 8x - 9$. Find:

a $\frac{dy}{dx}$

b The value of $\frac{dy}{dx}$ when $x = 3$

c $\frac{d^2y}{dx^2}$

d The value of $\frac{d^2y}{dx^2}$ when $x = 4$

e $\frac{d^3y}{dx^3}$

f The value of $\frac{d^3y}{dx^3}$ when $x = 6$

4 Find the second derivative of the following functions:

a $y = x^4 - 6x^3 + 8x^2 - 10$

b $f(x) = x^{\frac{1}{7}}$

c $f(x) = (3x + 2)^4$

d $f(x) = \frac{4}{(x + 8)^2}$

e $y = \left(7 - 6x^2\right)^{\frac{1}{3}}$

f $y = 4\sqrt{6 - x}$

g $f(x) = x^2\left(x^3 + 3\right)$

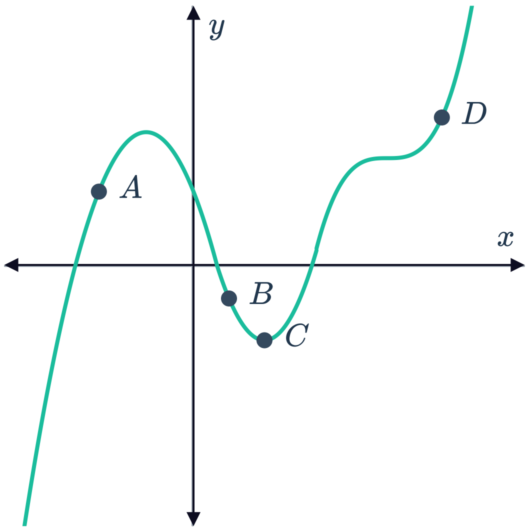
h $f(x) = \left(x^4 + 2\right)\left(3 - 2x^3\right)$

Concavity

5 Consider the given function $y = f(x)$ with points A, B, C and D:

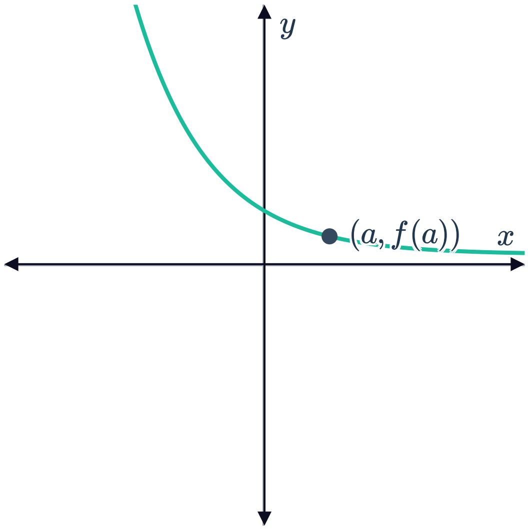
a At which of these points is $f''(x) > 0$?

b At which of these points is $f''(x) < 0$?



6 For each of the following curves, state the sign of $f'(a)$ and $f''(a)$:

a



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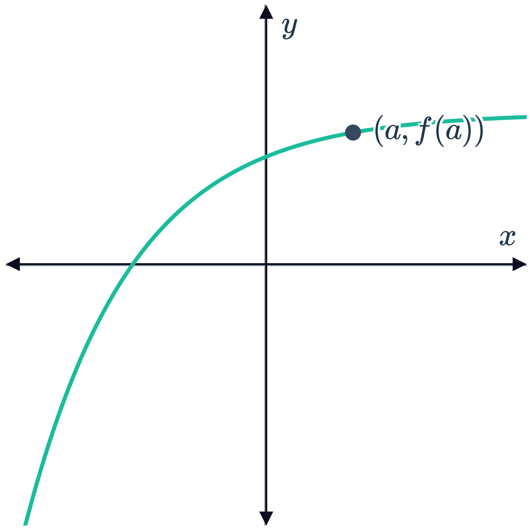

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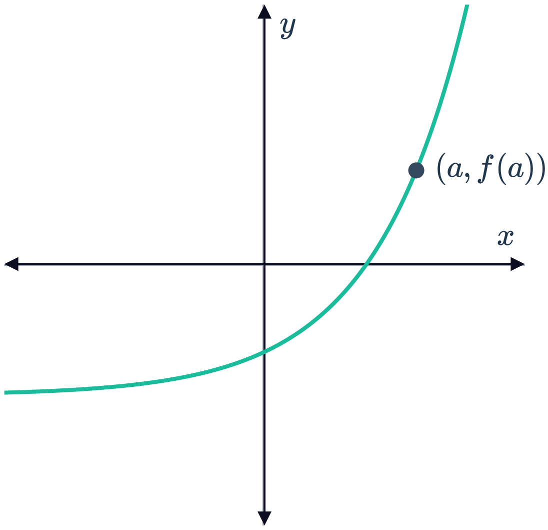

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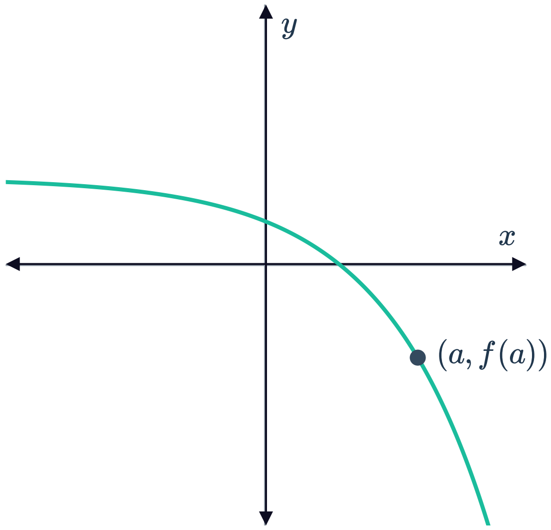

7 For each of the following quadratic functions:

i State whether the leading coefficient of $f(x)$ is positive or negative.

ii Hence, determine the nature of the turning point.

iii Find $f'(x)$.

iv Find $f''(x)$.

v State whether the curve is concave up or down.

a $f(x) = x^2 - 4x + 9$

b $f(x) = -x^2 + 4x - 9$

8 Consider the function $f(x) = 4x^2 + 3x + 2$.

a Find the first derivative.

b Find the second derivative.

c Complete the following tables of values:

x	0	1	2	3	4
$f'(x)$	3				

d Describe the rate of change of the gradient.

e Are there any points of inflection on $f(x)$? Explain your answer.

9 Consider the function $f(x) = -3x^2 + 12x + 2$.

a Find $f'(x)$.

b Find $f''(x)$.

c Describe the rate of change of the gradient.

d Are there any points of inflection on $f(x)$? Explain your answer.

10 For each of the following functions:

i State the values of x for which the graph of the function is concave up.

ii State the values of x for which the graph of the function is concave down.

iii Explain why there is no point of inflection.

a $y = \sqrt{x+1}$

b $y = -\sqrt{x-2}$

c $y = \dfrac{1}{x}$

d $y = \dfrac{1}{x^2}$

11 Consider the function $y=(x+6)^3$.

a State the transformation that turns $y=x^3$ into $y=(x+6)^3$.

b Find the point of inflection of $y=x^3$.

c Find the point of inflection of $y=(x+6)^3$.

d Complete the following table of values:

x	-7	-6	-5
y'			
y''		0	

e Is the point of inflection, a horizontal point of inflection? Explain your answer.

f For what values of x is the graph concave up?

12 Consider the function $y=(x+5)^3$.

a State the transformation that turns $y=x^3$ into $y=(x+5)^3$.

b Find the point of inflection of $y=x^3$

c Find the point of inflection of $y=(x+5)^3$.

d Complete the following table of values:

x	-6	-5	-4
y'			
y''		0	

e Is the point of inflection, a horizontal point of inflection? Explain your answer.

f For what values of x is the graph concave up?

13 Consider the function $y=x^4-8x^3-9$.

a Find y''.

b Find the points of inflection.

c Complete the following table of values:

x	-2	0	2	4	6
y'					
y''		0		0	

d Classify each point of inflection as an ordinary or horizontal point of inflection.

e For what values of x is the graph concave up?

14 Consider the function $y=4x^3-16x^2+4x+6$.

a Find y''.

b Find the point of inflection.

c Is the point of inflection, a horizontal or an ordinary point of inflection?

d For what values of x is the graph concave down?

15 Consider the function $y = 3x^3 + 6x^2 + 8x$

a Find y''.

b Find the point of inflection.

c Is the point of inflection, an ordinary or horizontal point of inflection?

d For what values of x is the graph concave down?

16 Consider the function $y = x^5 - 3x^2$.

a Find y''.

b Find the exact point of inflection.

c Is the point of inflection an ordinary or a horizontal point of inflection?

d For what values of x is the graph concave up?

e For what values of x is the graph concave down?

17 The first derivative of a certain function is $f'(x) = 3x^2 + 9x$.

a Determine the interval over which the function is increasing.

b Determine the interval over which the function is decreasing.

c Find $f''(x)$.

d Determine the interval over which the function is concave up.

e Determine the interval over which the function is concave down.

f Find the x-coordinate of the maximum turning point.

g Find the x-coordinate of the potential point of inflection.

Stationary points

18 Consider the function $f(x) = (x-5)^2 + 3$.

a Find the x-coordinate of the turning point.

b Find the value of the second derivative at this point.

c Hence, classify the stationary point.

19 Consider the function $f(x) = (x-5)(x+4)^2$.

a Find the turning points.

b Find the second derivative.

c Determine the nature of the turning points.

20 Consider the function $y = x^4 + 4x^3 + 2$.

a Find y'.

b Find y''.

c Find the stationary points.

d Classify the stationary points.

e Find the points of inflection.

f Classify each point of inflection as ordinary or horizontal point of inflection.

g For what values of x is the graph concave up?

21 Consider the function $f\left(x\right) = \left(x-8\right)\left(x-5\right)^2$.

a Find $f'\left(x\right)$.

b Find the turning points.

c Find $f''\left(x\right)$.

d Classify the turning points.

e For what values of x is the graph concave down?

22 For each of the following functions:

i Find the turning points.

ii Classify the turning points.

a $f(x) = x^3 - 27x - 7$

b $f(x) = x^3 - 12x - 2$

c $f(x) = 2x^3 + 9x^2 - 24x$

d $f(x) = x^4 - 2x^2 + 3$

23 For each of the following functions:

i Find the x-coordinates of the stationary points.

ii Determine the nature of the stationary points.

a $f\left(x\right) = 7x^3 + 8x^2 + 3x + 4$

b $f\left(x\right) = 8\left(x-7\right)^2\left(x+2\right)$

c $f\left(x\right) = -\dfrac{8}{x^2} + 2x$

d $f\left(x\right) = 2x^3 - 30x^2 + 150x - 247$

e $f\left(x\right) = 2\left(x-5\right)^3 + 1$

24 Consider the function $f\left(x\right) = 5x^3 + 7x^2 + 3x + 6$.

a Find the x-coordinates of stationary points.

b Find the x-value of the point of inflection.

25 Consider the function $y = \left(x^2 - 5\right)^3$.

a Find y''.

b Find the points of inflection.

c Which of these points are horizontal points of inflection?

26 For each of the following functions:

i Find the stationary point.

ii Find the possible point of inflection.

iii Determine whether this point is a turning point or a point of inflection. Explain your answer.

a $y = 4 - (x - 5)^4$ b $y = 5 - (x - 4)^4$ c $y = 6 - (x - 8)^4$

27 The function $f(x) = ax^2 + \dfrac{b}{x^2}$ has turning points at $x = 1$ and $x = -1$.

a Use the fact that there is a turning point at $x = 1$ to form an equation for a in terms of b.

b Use the fact that there is a turning point at $x = -1$ to form an equation for a in terms of b.

c What can you deduce about the values of a and b?

28 The function $f(x) = ax^3 + bx^2 + 12x + 5$ has a horizontal point of inflection at $x = 1$.

a Use information about $f'(x)$ to write an equation involving a and b.

b Use information about $f''(x)$ to write an equation for b in terms of a.

c Hence, find the value of a.

d Hence, find b.

29 The function $f(x) = ax^3 + 18x^2 + cx + 4$ has turning points at $x = 4$ and $x = 2$. Find the value of:

a a b c